

Project Work

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This document is the reviewed project work for the course Clustered Data Models.

Changes made are

- changed the type of Phase to factor
- the first box plot has been changed to a more appropriate one
- added modeling for variance and removed incorrect interpretation of variance
- conclusions using residual deviance have been removed
- added binned plots
- removed models, which did not work well with Phase as factorial.

The dataset

The dataset I chose to use for this project is *quilpie* from the GLMsData library in R. It describes the total July rainfall at Quilpie, Queensland, Australia from 1921 to 1988. There are 68 observations with 6 variables, of which I will be using the following 4:

- *Year* the year of the observation, one taken each year
- *SOI* the July average southern oscillation index, or SOI, a numeric value
- *Phase* the SOI phase; a factor with these values: 1 (consistently negative), 2 (consistently positive), 3 (rapidly falling), 4 (rapidly rising), or 5 (consistently near zero)
- *y* an indicator for whether or not the total monthly July rainfall exceeds 10 millimetres: a factor where 1 means the rainfall exceeds 10mm, and 0 means the rainfall is 10mm or less.

I realized a little bit too late, that a dataset with a hierarchical structure or even just more variables would've been more interesting to model but I did my best to make it work.

The research problem

In this project, I want to see, whether or not the SOI value and SOI phase can be used to model if the rainfall will exceed 10 millimetres or not. Phase is a categorical variable, that can be used for clustering in the models. I will be trying out a few different options and determining the best one out of them.

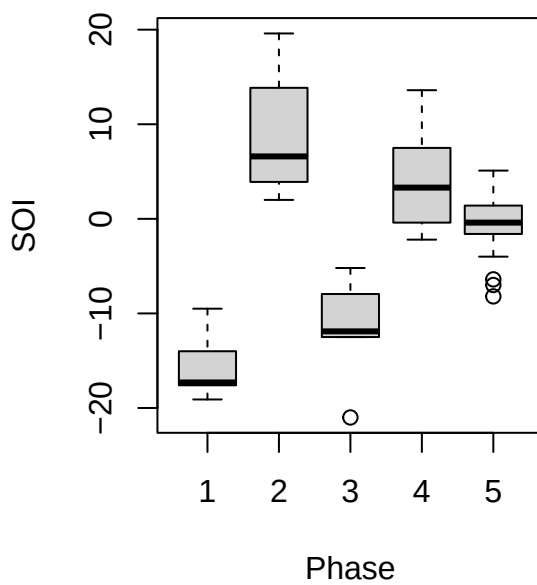
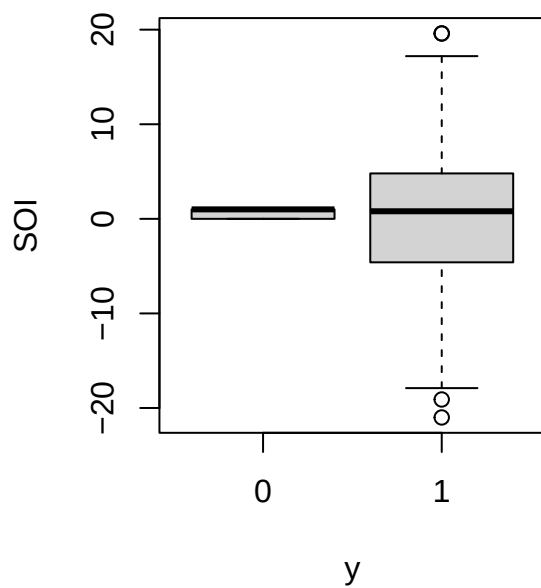
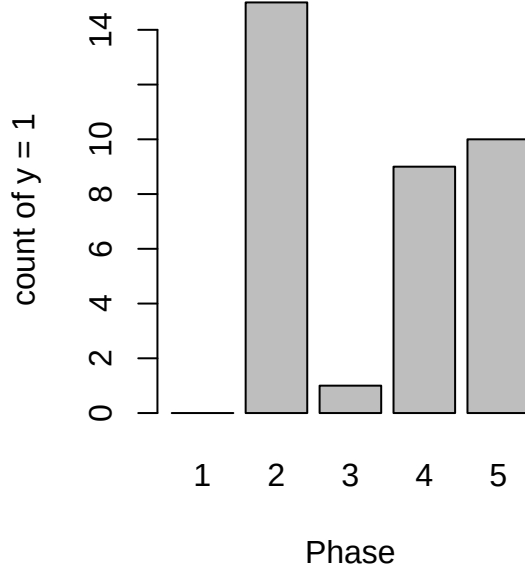
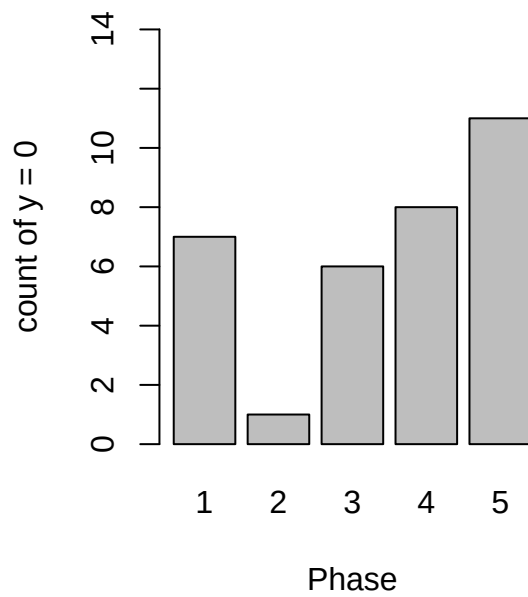
Visualizing the data

Before deciding, what kind of model will be appropriate, let's look at the variables available to us. Below is a figure with *SOI* and *Phase* plotted against the response variable *y*. It is apparent that there are differences in the values of both *Phase* and *SOI* for different *y* values. This means that there could be a possibility to model *y* using these two variables in some way.

Phase can have values 1, 2, 3, 4, and 5. There seems to be a difference in the distribution of phases between $y = 0$ and $y = 1$.

SOI is a numeric value between -20 and 20. When $y = 0$, *SOI* is around 0. When $y = 1$, *SOI* has a much larger variability and gets values on the whole -20 to 20 spectrum.

Below is also a graph with *SOI* and *Phase* plotted against $y = 0$ and $y = 1$. It can be seen that neither *SOI* nor *Phase* are constant between the different values of *y*. We can also notice that values of *SOI* seem to be different depending on the value of *Phase*.



Modeling the variance in the case of binomial data, where the response is the number of successes (or a proportion of successes) for different combinations of values of explanatory variables. The model is overfitted and the results do not make much sense.

```
##
## Call:
```

```
## glm(formula = cbind(successes, failures) ~ factor(Phase), family = binomial(),
##      data = data.frame(bin_dat))
##
## Deviance Residuals:
## [1]  0  0  0  0  0
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -24.81   55959.15      0      1
## factor(Phase)2    27.52   55959.15      0      1
## factor(Phase)3    23.02   55959.15      0      1
## factor(Phase)4    24.93   55959.15      0      1
## factor(Phase)5    24.72   55959.15      0      1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 2.8414e+01  on 4  degrees of freedom
## Residual deviance: 2.3496e-10  on 0  degrees of freedom
## AIC: 20.615
##
## Number of Fisher Scoring iterations: 22
```

Modeling

Generalized Linear Model

Since the response variable y is binomial, it makes sense to start with a simple logistic GLM with SOI as the explanatory variable. The following is the summary of this model.

```
##
## Call:
## glm(formula = y ~ SOI, family = binomial)
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -2.0793  -0.9853   0.3245   0.9616   1.9303
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  0.04813    0.27856   0.173 0.862828
## SOI          0.14642    0.04288   3.415 0.000638 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 94.209  on 67  degrees of freedom
## Residual deviance: 75.901  on 66  degrees of freedom
## AIC: 79.901
##
## Number of Fisher Scoring iterations: 4
```

This model has an intercept of 0.05 and SOI coefficient of 0.15, with SOI being a statistically significant variable with the biggest possible confidence interval. The standard errors are 0.28 and 0.04 for the intercept and SOI respectively. The AIC value is 79.901, which will become relevant later on, when comparing different models.

Generalized Linear Model with interaction

Next I decided to try adding *SOI* and *Phase* interaction variable into the previous model. This resulted in the following logistic GLM.

```
##
## Call:
## glm(formula = y ~ SOI * Phase, family = binomial)
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -2.2453  -1.1243   0.1205   1.1052   1.9683
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -1.757e+01  7.745e+03  -0.002   0.998
## SOI          -5.269e-09  4.876e+02   0.000   1.000
```

```
## Phase2      1.913e+01  7.745e+03  0.002  0.998
## Phase3      1.545e+01  7.745e+03  0.002  0.998
## Phase4      1.777e+01  7.745e+03  0.002  0.998
## Phase5      1.748e+01  7.745e+03  0.002  0.998
## SOI:Phase2   1.714e-01  4.876e+02  0.000  1.000
## SOI:Phase3  -2.844e-02  4.876e+02  0.000  1.000
## SOI:Phase4  -2.357e-02  4.876e+02  0.000  1.000
## SOI:Phase5   1.104e-02  4.876e+02  0.000  1.000
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 94.209  on 67  degrees of freedom
## Residual deviance: 65.204  on 58  degrees of freedom
## AIC: 85.204
##
## Number of Fisher Scoring iterations: 16
```

This model does not have any statistically significant variables and the AIC is 85.204. This indicates a worse fit than the first model.

Generalized Linear Mixed Model

As there is variability in the baseline level of y across different phases (groups), it makes sense to try adding *Phase* as a random effect. This creates a logistic GLMM with *SOI* as the explanatory variable and *Phase* as the random effect, and a summary of it is like follows.

```
## boundary (singular) fit: see help('isSingular')

## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##   Family: binomial ( logit )
## Formula: y ~ SOI + (1 | Phase)
##
##      AIC      BIC    logLik deviance df.resid
##      81.9      88.6     -38.0     75.9      65
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.7724 -0.7912  0.2325  0.7667  2.3329
##
## Random effects:
##   Groups Name            Variance Std.Dev.
##   Phase (Intercept) 0          0
## Number of obs: 68, groups: Phase, 5
##
## Fixed effects:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  0.04813    0.27856   0.173 0.862829
## SOI          0.14642    0.04288   3.415 0.000638 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Correlation of Fixed Effects:
##      (Intr)
## SOI -0.056
## optimizer (Nelder_Mead) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

The random effect for Phase shows no variance (0), suggesting that there is no meaningful random variability across the different phases, so a GLMM might not be the best choice.

Model comparisons

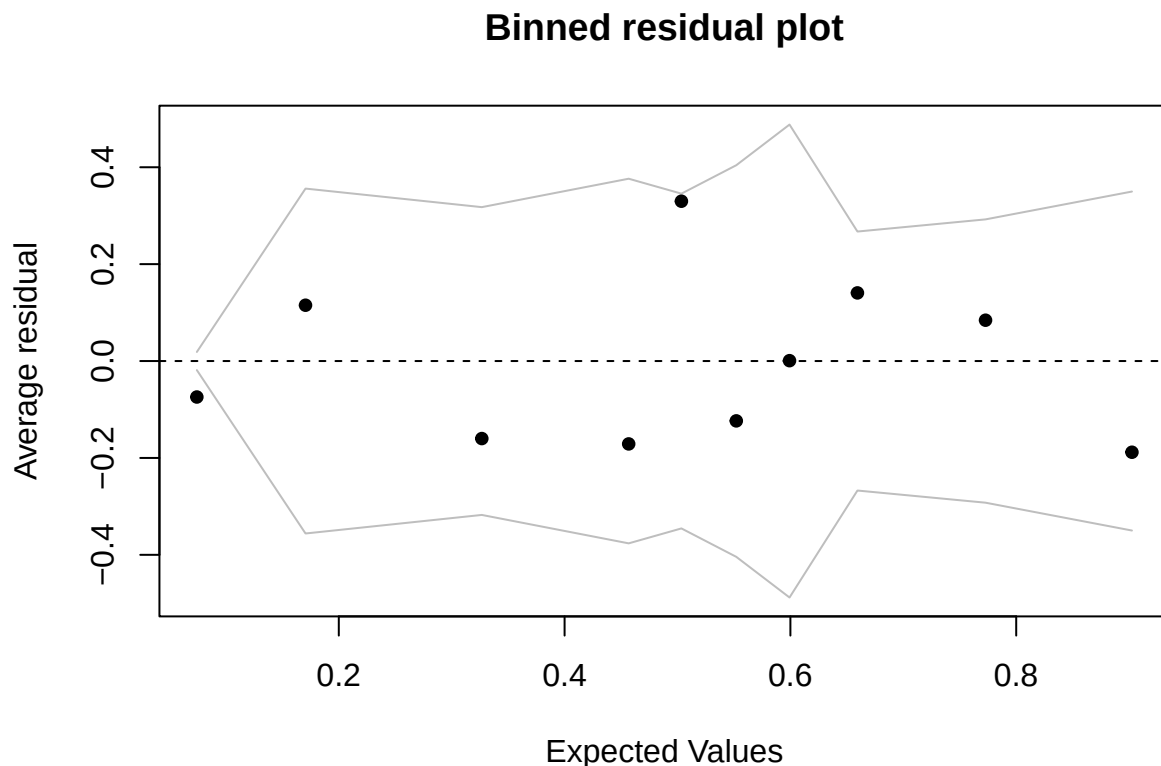
Now that we have tried out multiple models, let's summarize their comparison. Firstly, let's look at the AIC values for the simple and interaction logistic GLMs and the GLMM model.

```
##          df      AIC
## glm_mod   2 79.9006
## glm_mod2 10 85.2043
## glmm_mod  3 81.9006
```

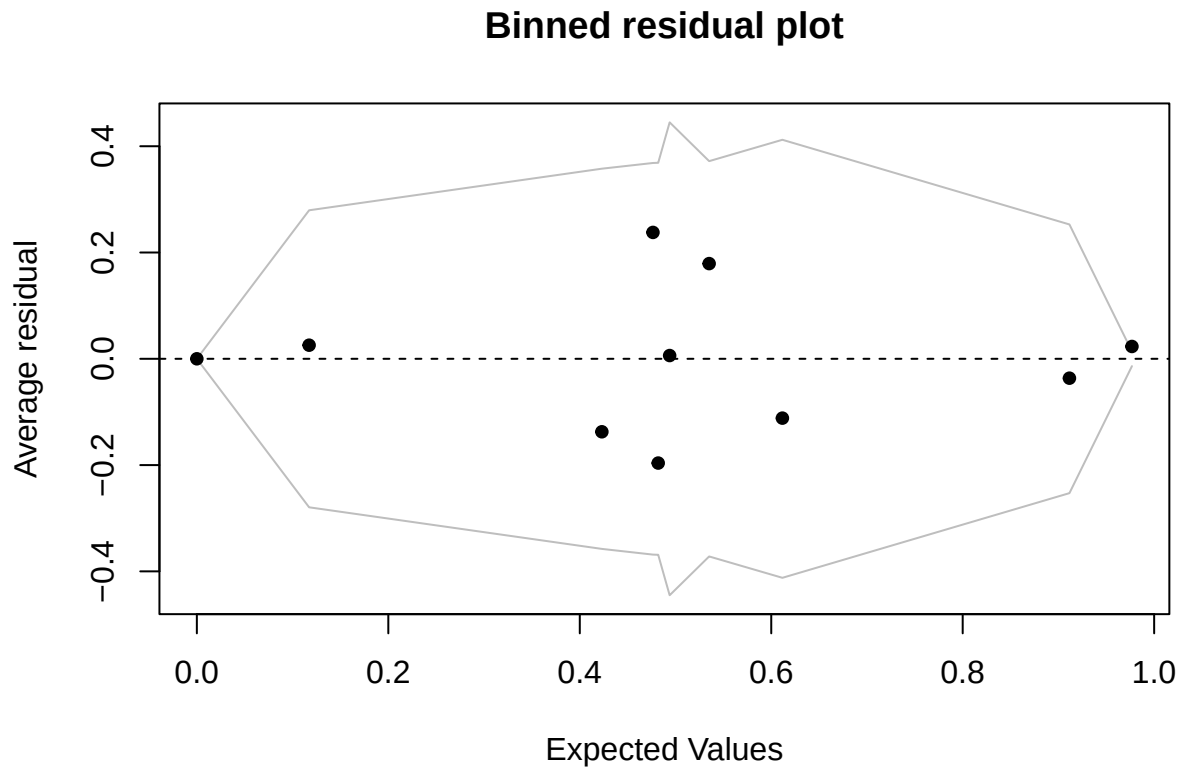
The best one according to AIC values is the simplest GLM model.

Model fit of the best model

Let's look at the residuals of the models.



The residuals look good for the first and simplest GLM model. There are no clear outliers and there is no distinguishable pattern. Most binned residuals fall into the expected interval.



There is a visible pattern in the second GLM model. Even though residuals fall into the expected area, they do not look random. This also suggests that the simpler model is better in this case.

Based on the information above, I chose first GLM model as the best fit. The residuals below look good and the AIC value was the smallest of the three models compared.

```
##
## Call:
## glm(formula = y ~ SOI, family = binomial)
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -2.0793  -0.9853   0.3245   0.9616   1.9303
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  0.04813    0.27856   0.173  0.862828
## SOI          0.14642    0.04288   3.415  0.000638 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 94.209  on 67  degrees of freedom
## Residual deviance: 75.901  on 66  degrees of freedom
## AIC: 79.901
##
```

Number of Fisher Scoring iterations: 4

Formula of the model

Now let's dissect the GLM model that I ended up with. The model formula is

$$\text{logit}\left(\frac{P(y = 1)}{1 - P(y = 1)}\right) = \beta_0 + \beta_1 \cdot \text{SOI}$$

where y is the binary response variable,

β_0 is the intercept and

β_1 is the coefficient for the variable SOI .

The results and interpretation

Now we can deduct that

- For each 1-unit increase in SOI, the log-odds of the total July rainfall at Quilpie, Queensland, Australia from 1921 to 1988 being over 10 mm increase by 0.14642
- SOI is a statistically significant variable and there is no visible random effect from Phase.

Conclusion

In the case of the quilpie dataset in GLMsData in R, the best model to describe the binomial variable y (whether or not rainfall exceeds 10 mm), according to my explorations and tests, is the simple GLM model with SOI as the only explanatory variable. I am certain the model could be made better but I had trouble getting the more complicated models to work or be able to compare them to the three models that I explored in this project.

Overall, I think the model I chose works well for this data and models the log-odds of rainfall exceeding 10 mm at Quilpie, Queensland, Australia from 1921 to 1988 sufficiently. I think there could have been more statistics and diagnostics to compare the models but I found it quite difficult due to the significant differences between the models and their assumptions.

Appendix

```
# libraries
library(GLMsData)
library(glmmML)
library(lme4)
library(nlme)
library(gee)
library(dplyr)
library(arm)

# getting the data
data(quilpie)

# changing Phase to factor
quilpie$Phase <- factor(quilpie$Phase)
attach(quilpie)

# plots
par(mfrow = c(1, 2))
y0 <- subset(quilpie, y == 0)
y1 <- subset(quilpie, y == 1)
barplot(table(y0$Phase), xlab="Phase", ylab="count of y = 0", ylim=c(0,14))
barplot(table(y1$Phase), xlab="Phase", ylab="count of y = 1")
par(mfrow = c(1, 2))
boxplot(y, SOI, ylab= "SOI", xlab = "y", names = c(0,1))
plot(Phase, SOI, ylab= "SOI", xlab = "Phase", names = c(1,2,3,4,5))

# modeling variance
trials <- 0
successes <- 0
for (x in 1:5) {
  s = subset(quilpie,Phase==x)
  successes <- append(successes, sum(s$y))
  trials <- append(trials, nrow(s))
}
bin_dat <- cbind(successes=successes, failures = trials-successes,
                 Phase= c(0,1,2,3,4,5))[2:6,]

variance_mod <- glm(cbind(successes, failures) ~ factor(Phase),
                   family = binomial(), data = data.frame(bin_dat))
summary(variance_mod)

# first model
glm_mod <- glm(y ~ SOI, family=binomial)
summary(glm_mod)
#plot(residuals(glm_mod, type = "pearson")) +
#  abline(h=0)

# second model
glm_mod2 <- glm(y ~ SOI*Phase, family=binomial)
summary(glm_mod2)
#plot(residuals(glm_mod2, type = "pearson")) +
```

```

# abline(h=0)

# third model
glmm_mod <- glmer(y ~ SOI + (1|Phase), family = binomial)
summary(glmm_mod)
#plot(residuals(glmm_mod, type = "pearson")) +
#abline(h=0)

# AIC for all three models
AIC(glm_mod, glm_mod2, glmm_mod)

# binned plots for first two models
par(mfrow = c(1, 1))
binnedplot(fitted(glm_mod), residuals(glm_mod, type = "response"))

binnedplot(fitted(glm_mod2), residuals(glm_mod2, type = "response"))

# best fit
summary(glm_mod)

```